

2. We will need to use the following command to ensure that the required Perl modules are installed:

```
cpan -i module-name
```

Here, *module-name* needs to be replaced with *Date::Calc*, *Date::HiRes*, and *Date::Format*.

Download the tar file from the Sourceforge website and unzip it.

Use the following command to extract the files:

```
Star -zxvf swatch-3.2.3.tar.gz
```

3. Since Swatch is a Perl program, the installation process is slightly different. Here is the sequence of commands that one needs to follow:

```
perl Makefile.PL
make
make test
make install
make realclean
```

Once these processes are done, Swatch is installed and we are ready to go.

After installing Swatch, one might be interested in creating a configuration file. If you look at the contents of a Swatch configuration file, you will notice that the syntax is very simple – all it requires is a definition of what to search for, followed by an action or remedy if a specific match is located. It is important to note here that Swatch uses Perl regular expressions as a way to define the search parameters and can perform a variety of actions.

First, you need to create an empty file to be used as a configuration file. The normal practice is to create this file under */etc*. One can modify the configuration file by using any basic text editor. The Swatch configuration file is where all the important settings are located. In this file, called *swatchrc* by default, one can instruct the program what to look for in the log files and what to do if what has been specified does show up. Since the whole point of Swatch is to simplify our lives, configuring Swatch is pretty simple. As stated earlier, Swatch is controlled by a single file, *\$HOME/.swatchrc* (by default). This file contains text patterns, in the form of regular expressions that you want Swatch to watch for. Each regular expression is followed by the action(s) you wish Swatch to take whenever it encounters that text.

The configuration file syntax begins with a ‘watchfor’ keyword. Basically, this line instructs Swatch to search for specific patterns in the form of regular expressions. Each ‘watchfor’ line is then preceded by an action.

There are two options available in *swatchrc* to specify the patterns to watch out for. These are:

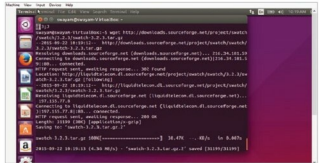


Figure 1: Download the file from SourceForge site

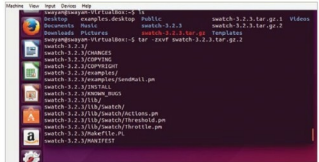


Figure 2: Files are extracted and ready for installation

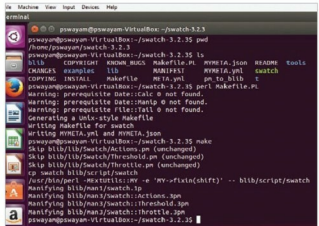


Figure 3: Execution of *make* commands

1. **Watch for regex** – Here, an appropriate action will be taken when the regular expression specified in *regex* is found within the file or command being monitored.
2. **Ignore regex** – Take an action when there is any activity within the file or command being monitored, except for events that match the expression specified in *regex*.

In the following example, we will simply output the log entry to the console if an *ssh* connection has been established by issuing the *echo* action.

```
watchfor /ssh/
echo bold
```

Let us take a quick look at some of the options that are